Year 11 Mathematics

Algebra 7

2.2 Change of Subject

Change of Subject

STARTER · 5 MIN

Date: _____ **Title:** Change of Subject

LEARNING OBJECTIVES

- Rearrange simple formulae
- Rearrange formulae with brackets, fractions and powers
- Use rearranged formulae to solve problems

STARTER *Answer in your book. No calculators.*

1. Solve $3x + 5 = 20$
2. Solve $\frac{x}{4} = 7$
3. Solve $2(x - 3) = 10$
4. Solve $x^2 = 49$

Starter — Answers

STARTER

Tick or correct in green pen

1. $3x = 15, \quad x = \mathbf{5}$

2. $x = \mathbf{28}$

3. $x - 3 = 5, \quad x = \mathbf{8}$

4. $x = \mathbf{\pm 7}$

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can rearrange simple formulae	☐
SILVER	I can rearrange formulae with brackets, fractions and powers	☐
GOLD	I can use rearranged formulae to solve problems	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Change of Subject

INTRODUCTION · 3 MIN

The subject is the letter on its own. Use inverse operations, like solving an equation.

Inverse operations

Undo + with −, × with ÷.

$y = x + 5 \rightarrow x = y - 5$

Two steps

Undo the last operation first.

$y = 3x - 2 \rightarrow x = (y + 2) \div 3$

Fractions

Multiply to clear the fraction.

$y = \frac{x}{4} \rightarrow x = 4y$

Powers and roots

Undo a square with a square root.

$A = x^2 \rightarrow x = \sqrt{A}$

Examples 1 and 2 — Simple Rearranging

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Example 1

Make x the subject of

$y = 4x + 7$

Example 2

Make a the subject of

$v = u + at$

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

$$y - 7 = 4x$$

$$x = \frac{y - 7}{4}$$

Tip: subtract 7, then divide by 4.

Example 2

$$v - u = at$$

$$a = \frac{v - u}{t}$$

Tip: treat other letters like numbers.

Task 1

PRACTICE · 9 MIN

BRONZE *Make x the subject*

1. $y = x + 9$
2. $y = 5x$
3. $y = 2x - 3$
4. $p = 6 + 3x$
5. $y = mx + c$

★ $A = x - 4b$

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1 $x = y - 9$

2 $x = \frac{y}{5}$

3 $x = \frac{y + 3}{2}$

4 $x = \frac{p - 6}{3}$

5 $x = \frac{y - c}{m}$

★ $x = A + 4b$

Examples 3 and 4 — Brackets, Fractions and Powers

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Example 3

Make x the subject of

$y = 3(x + 2)$

Example 4

Make r the subject of

$A = \pi r^2$

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

$$\frac{y}{3} = x + 2$$

$$x = \frac{y}{3} - 2$$

Tip: divide by 3 first.

Example 4

$$\frac{A}{\pi} = r^2$$

$$r = \sqrt{\frac{A}{\pi}}$$

Tip: undo the square with a square root.

Task 2

PRACTICE · 9 MIN

SILVER

Make the letter in brackets the subject

1. $y = \frac{x}{3} + 2 \quad (x)$

2. $P = 2(l + w) \quad (l)$

3. $V = x^3 \quad (x)$

4. $y = \frac{2x - 1}{5} \quad (x)$

5. $E = mc^2 \quad (c)$

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1 $x = 3(y - 2)$

2 $l = \frac{P}{2} - w$

3 $x = \sqrt[3]{V}$

4 $x = \frac{5y + 1}{2}$

5 $c = \sqrt{\frac{E}{m}}$

Examples 5 and 6 — Harder Rearranging

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Example 5 (2 marks)

Make t the subject of

$s = \frac{1}{2}at^2$

Example 6 (2 marks)

Make u the subject of

$v^2 = u^2 + 2as$

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

$$2s = at^2$$

$$\frac{2s}{a} = t^2$$

$$t = \sqrt{\frac{2s}{a}}$$

Tip: clear the half first.

Example 6

$$v^2 - 2as = u^2$$

$$u = \sqrt{v^2 - 2as}$$

Tip: square root the whole expression.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: show all your working

1. Make h the subject of $V = \pi r^2 h$ (2 marks)

2. Make x the subject of $y = \sqrt{x + 3}$ (2 marks)

3. Make C the subject of $F = 1.8C + 32$ (2 marks)

4. Make r the subject of $V = \frac{4}{3}\pi r^3$ (3 marks)

5. Make l the subject of $T = 2\pi\sqrt{l/g}$ (3 marks)

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1 $h = \frac{V}{\pi r^2}$

2 $x = y^2 - 3$

3 $C = \frac{F - 32}{1.8}$

4 $r = \sqrt[3]{\frac{3V}{4\pi}}$

5 $I = \frac{gT^2}{4\pi^2}$

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Make x the subject of $y = x - 8$

SILVER Make x the subject of $y = 5x + 1$

GOLD Make r the subject of $C = 2\pi r$

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE $x = y + 8$

SILVER $x = \frac{y - 1}{5}$

GOLD $r = \frac{C}{2\pi}$

Exam Practice — Simple Rearranging

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: answer on the line, show all your working

Question 1 (2 marks)

Make x the subject of

$y = 5x - 3$

.....

Question 2 (2 marks)

Make t the subject of

$v = u + at$

.....

Exam Practice — Brackets, Fractions and Powers

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: answer on the line, show all your working

Question 3

(2 marks)

Make x the subject of

$$y = \frac{2x - 1}{5}$$

.....

Question 4

(2 marks)

Make r the subject of

$A = \pi r^2$

.....

Exam Practice — Harder Rearranging

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: answer on the line, show all your working

Question 5 (2 marks)

Make u the subject of

$v^2 = u^2 + 2as$

.....

Question 6 (3 marks)

Make l the subject of

$$T = 2\pi\sqrt{\frac{l}{g}}$$

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

$y + 3 = 5x$ — M1

$x = \dfrac{y + 3}{5}$ — A1

Tip: undo − 3 first, then ÷ 5.

Question 2

$v - u = at$ — M1

$t = \dfrac{v - u}{a}$ — A1

Tip: treat the other letters like numbers.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

$5y = 2x - 1$ **M1**

$\mathbf{x = \frac{5y + 1}{2}}$ **A1**

Tip: multiply by 5 to clear the fraction.

Question 4

$r^2 = \frac{A}{\pi}$ **M1**

$\mathbf{r = \sqrt{\frac{A}{\pi}}}$ **A1**

Tip: square root the whole fraction.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

$u^2 = v^2 - 2as$ — M1

$\mathbf{u = \sqrt{v^2 - 2as}}$ — A1

Tip: the root goes over everything.

Question 6

Divide by 2π, then square:

$$\frac{T^2}{4\pi^2} = \frac{l}{g}$$ — M1 M1

$$\mathbf{l = \frac{gT^2}{4\pi^2}}$$ — A1

Tip: divide by 2π, then square both sides.